CLARIN Federated Content Search (CLARIN-FCS) - Data Views 1.0

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Chapter 1. Introduction

This specification is a supplementary specification to the CLARIN-FCS Core specification and defines additional Data View to be used in CLARIN-FCS. This specification defines the supplementary Data Views. For detailed information about the CLARIN-FCS interface specification, see CLARIN-FCS-Core 2.0.

1.1. Terminology

The key words `MUST`, `MUST NOT`, `REQUIRED`, `SHALL`, `SHALL NOT`, `SHOULD`, `SHOULD NOT`, `RECOMMENDED`, `MAY`, and `OPTIONAL` in this document are to be interpreted as in RFC2119.

1.2. Normative References

RFC2119

Key words for use in RFCs to Indicate Requirement Levels, IETF RFC 2119, March 1997, https://www.ietf.org/rfc/rfc2119.html

XML-Namespaces

Namespaces in XML 1.0 (Third Edition), W3C, 8 December 2009, http://www.w3.org/TR/2009/REC-xml-names-20091208/

CLARIN-FCS-Core 2.0

CLARIN Federated Content Search (CLARIN-FCS) - Core 2.0, SCCTC FCS Task-Force, May 2017, https://trac.clarin.eu/wiki/FCS/Specification, https://office.clarin.eu/v/CE-2017-1046-FCS-Specification-v89.pdf

1.3. Non-Normative References

RFC6838

Media Type Specifications and Registration Procedures, IETF RFC 6838, January 2013, https://www.ietf.org/rfc/rfc6838.html

RFC3023

XML Media Types, IETF RFC 3023, January 2001, https://www.ietf.org/rfc/rfc3023.html

KML

Keyhole Markup Language (KML), Open Geospatial Consortium, 2008, https://www.opengeospatial.org/standards/kml

1.4. Typographic and XML Namespace conventions

The following typographic conventions for XML fragments will be used throughout this specification:

- `<prefix:Element>`

An XML element with the Generic Identifier *Element* that is bound to an XML namespace denoted by the prefix *prefix*.

- `@attr`

  An XML attribute with the name *attr*.

- `string`

  The literal *string* must be used either as element content or attribute value.

Endpoints and Clients `MUST` adhere to the XML-Namespaces specification. The CLARIN-FCS interface specification generally does not dictate whether XML elements should be serialized in their prefixed or non-prefixed syntax, but Endpoints `MUST` ensure that the correct XML namespace is used for elements and that XML namespaces are declared correctly. Clients `MUST` be agnostic regarding syntax for serializing the XML elements, i.e. if the prefixed or un-prefixed variant was used, and `SHOULD` operate solely on expanded names, i.e. pairs of namespace name and local name.

The following XML namespace names and prefixes are used throughout this specification. The column "Recommended Syntax" indicates which syntax variant `SHOULD` be used by the Endpoint to serialize the XML response.

Table 1. XML Namespaces and prefixes

Prefix	Namespace Name	Comment	Recommended Syntax
`fcs`	`http://clarin.eu/fcs/resource`	CLARIN-FCS Resources	prefixed
`cmdi`	`http://www.clarin.eu/cmd/`	Component Metadata	un-prefixed
`kml`	`http://www.opengis.net/kml/2.2`	Keyhole Markup Language	un-prefixed

Chapter 2. Data Views

A Data View serves as a container for representing search results within CLARIN-FCS. Data Views are designed to allow for different representations of results. This specification defines supplementary Data Views beyond the Generic Hits Data View and Advanced Data View, that is defined as part of the CLARIN-FCS Core 2.0 specification. For detailed information as to what Data Views are and how they are integrated in CLARIN-FCS, see CLARIN-FCS-Core 2.0.

NOTE The examples in the following sections show only the payload with the enclosing `<fcs:DataView>` element of a Data View. Of course, the Data View must be embedded either in a `<fcs:Resource>` or a `<fcs:ResourceFragment>` element. The `@pid` and `@ref` attributes have been omitted for all inline payload types.

2.1. Generic Hits (HITS)

The *Generic Hits* (HITS) Data View is an integral part of the Core specification and serves as the as the *most basic* agreement in CLARIN-FCS for the serialization of search results. For details about this Data View, see the Core specification CLARIN-FCS-Core 1.0 or 2.0, Section "Generic Hits (HITS)".

2.2. Advanced (ADV)

The *Advanced* (ADV) Data View is an integral part of the Core 2.0 specification and serves as the as the agreement in CLARIN-FCS for the serialization of CLARIN-FCS 2.0 Advanced Search results. For details about this Data View, see the Core specification CLARIN-FCS-Core 2.0, Section "Advanced (ADV)".

2.3. Component Metadata (CMDI)

Description	A CMDI metadata record
MIME type	`application/x-cmdi+xml`
Payload Disposition	*inline* or *reference*
Payload Delivery	*send-by-default* (`RECOMMENDED`)
Recommended Short Identifier	`cmdi` (`RECOMMENDED`)

The *Component Metadata Data View* allows the embedding of a CMDI metadata record that is *applicable* to the specific context into the Endpoint response, e.g. metadata about the resource in which the hit was produced. If this CMDI record is applicable for the entire Resource, it `SHOULD` be put in a `<fcs:DataView>` element below the `<fcs:Resource>` element. If it is applicable to the Resource Fragment, i.e. it contains more specialized metadata than the metadata for the encompassing resource, it `SHOULD` be put in a `<fcs:DataView>` element below the `<fcs:ResourceFragment>` element. Endpoints `SHOULD` provide the payload *inline*, but Endpoints `MAY` also use the *reference* method. If an Endpoint uses the *reference* method, the CMDI metadata record `MUST` be downloadable without any restrictions.

Example (inline)

```
<!-- potential @pid and @ref attributes omitted -->
<fcs:DataView type="application/x-cmdi+xml">
  <cmdi:CMD xmlns:cmdi="http://www.clarin.eu/cmd/" CMDVersion="1.1">
    <!-- content omitted -->
  </cmdi:CMD>
</fcs:DataView>
```

Example (referenced)

```
<!-- potential @pid attribute omitted -->
<fcs:DataView type="application/x-cmdi+xml"
ref="http://repos.example.org/resources/4711/0815.cmdi" />
```

2.4. Images (IMG)

Description	An image related to the hit
MIME type	`image/png`, `image/jpeg`, `image/gif`, `image/svg+xml`
Payload Disposition	*reference*
Payload Delivery	*need-to-request* (`RECOMMENDED`)
Recommended Short Identifier	`image` (`RECOMMENDED`)

The *Image Data View* provides an image that is relevant to the hit, e.g. a facsimile of the source of a transcription. Endpoints `MUST` provide the payload by the *reference* method and the image file `SHOULD` be downloadable without any restrictions.

Example

```
<!-- potential @pid attribute omitted -->
<fcs:DataView type="image/png" ref="http://repos.example.org/resources/4711/0815.png"
/>
```

2.5. Geolocation (GEO)

Description	An geographic location related to the hit
MIME type	`application/vnd.google-earth.kml+xml`
Payload Disposition	*inline*
Payload Delivery	*need-to-request* (`RECOMMENDED`)
Recommended Short Identifier	`kml` (`RECOMMENDED`)

The *Geolocation Data View* allows to geolocalize a hit. If MUST be encoded using the XML representation of the Keyhole Markup Language (KML). The KML fragment MUST comply with the specification as defined by [ref:KML].

Example

```
<!-- potential @pid and @ref attributes omitted -->
<fcs:DataView type="application/vnd.google-earth.kml+xml">
  <kml:kml xmlns:kml="http://www.opengis.net/kml/2.2">
    <kml:Placemark>
      <kml:name>IDS Mannheim</kml:name>
      <kml:description>Institut für Deutsche Sprache, R5 6-13, 68161 Mannheim,
Germany</kml:description>
      <kml:Point>
        <kml:coordinates>8.4719510,49.4883700,0</kml:coordinates>
      </kml:Point>
    </kml:Placemark>
  </kml:kml>
</fcs:DataView>
```